Namrata Nadagouda

Email: namrata.nadagouda@gatech.edu
Website: https://nnadagouda95.github.io
Google Scholar: https://scholar.google.com/citations?user=WPOYaFAAAAAJ&hl=en

OVERVIEW

- Research on developing methods for learning data efficient models based on active learning, human-in-the-loop learning and learning from comparisons such as similarity judgments and user preferences
- Worked on applications such as preference learning, control parameter tuning, metric learning, and classification across images, text and time series data
- Have a well rounded knowledge of ML theory and practice including both traditional ML and deep learning methods
- Experience with Python, MATLAB, PyTorch, SQL, Labelbox, Azure, AWS

EDUCATION

Ph.D. Electrical & Computer Engineering *December 2026 (Expected)*
Machine Learning and Digital Signal Processing
Georgia Institute of Technology, Atlanta, GA
Advisor: Prof. Mark Davenport
Thesis: "Building data efficient models using active learning and similarity comparisons"

M.S. Electrical & Computer Engineering December 2020
Machine Learning and Digital Signal Processing
Georgia Institute of Technology, Atlanta, GA

B.Tech. Electrical & Electronics Engineering May 2017
Digital Signal Processing and Digital System Design
National Institute of Technology Karnataka, Surathkal, India

PUBLICATIONS

N. Nadagouda, N. Ahad, M. Tucker and M. Davenport, "Active query synthesis for learning from preferences", *Under review.* Also presented at Women in Machine Learning (WiML) Symposium, co-located with *International Conference on Machine Learning (ICML), July 2026.*

N. Nadagouda, A. Xu and M. Davenport, "Active metric learning and classification using similarity queries", in *Uncertainty in Artificial Intelligence (UAI)*, August 2023. Also presented at Human in the Loop Learning Workshop, *Neural Information Processing Systems (NeurIPS), December 2022.*

A. McRae, A. Xu, J. Jin, **N. Nadagouda**, N. Ahad, P. Guan, S. Karnik and M. Davenport, "Delta Distancing: A Lifting Approach to Localizing Items From User Comparisons", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), May 2022.*

N. Nadagouda and M. Davenport, "Switched Hawkes Processes", in *Proc. IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP), June 2021.*

G. Canal, M. Connor, J. Jin, **N. Nadagouda**, M. O'Shaughnessy, C. Rozell and M. Davenport, "The Picasso Algorithm for Bayesian Localization Via Paired Comparisons in a Union of Subspaces Model", in *Proc. IEEE Int. Conf. on Acoustics,*

Speech and Signal Processing (ICASSP), May 2020.

ABSTRACTS

N. Ahad, **N. Nadagouda**, E. Dyer and M. Davenport, "Active learning for time instant classification", at Data-centric Machine Learning Research Workshop, *International Conference on Machine Learning (ICML), July 2023.*

Y. Teng, A. Mamuye, E. Mo, K. Zhu, R. Walker, **N. Nadagouda** and M. Davenport, "Range-Only Simultaneous Localization and Mapping using Paired Comparisons", at *IEEE Annual Conf. on RFID, April 2021.*

N. Nadagouda and M. Davenport, "Switched Hawkes Processes", at *Workshop on Recent Developments on Mathematical/Statistical approaches in Data Science (MSDAS), June 2019.*

ARTICLES

N. Nadagouda, "Journey of a researcher: Finding pleasure in the pathless woods", *American Ceramic Society Bulletin*, Student Perspectives, June/July 2020.

WORK EXPERIENCE

Applied Scientist Intern Summer 2024
ML Science Team, Samsara, San Francisco, CA

- Worked on active learning based traffic sign detection model using trip still images from dashcams
- Built the infrastructure to actively select images for labeling using a pre-trained distilled model and fine-tune the model using the labels obtained
- Also, involved data preparation which comprised of text prompt based collection of data using a CLIP based similarity search tool and labeling using Labelbox
- Conducted initial experiments with varying batch sizes of acquired data and the findings were handed off to the team for further investigation

Intern Summer 2018
Hedge Fund Start-up, Atlanta, GA

- Worked on data management and data pre-processing of stock trade data stored in SQL databases.
- Also, involved performance evaluation of trading algorithms on Microsoft Azure platform

Research Intern, Microarchitecture Research Lab Fall 2016
Intel India - Intel Labs, Bangalore, India

Research Intern Summer 2016
Department of Electrical Communication Engineering
Indian Institute of Science, Bangalore, India

TALKS

Active query synthesis for preference learning 2026
CSIP Seminar, Georgia Tech, Atlanta, GA

Active query synthesis for preference learning 2025
ML@GT Student Conference, Georgia Tech, Atlanta, GA

AWM Workshop at SIAM Annual Meeting, Montreal, Canada

Active learning for traffic sign detection 2024
Samsara, San Francisco, CA

Building data efficient models using active learning and similarity comparisons 2022
Thesis presentation at Georgia Tech, Atlanta, GA

Switched Hawkes Processes 2019
Student Seminar at Georgia Tech, Atlanta, GA

AWARDS

2nd place, Presentations — September 2025
ML@GT Student Conference
Georgia Tech

Registration and Travel Awards

AWM Workshop at SIAM Annual Meeting — July 2025
Travel Funding

NeurIPS — December 2023
Conference Volunteer Financial Assistance
WiML Travel Funding

Women in Data Science and Mathematics — August 2023
Institute for Pure and Applied Mathematics
Workshop Financial Assistance
GT Career Development Funding

UAI — August 2023
Conference Funding

ICML — August 2023
GT SGA Conference Funding
CoE Professional Development Funding
WiML Funding

NeurIPS — December 2022
GT SGA Conference Funding
CoE Professional Development Funding

Deep Learning Theory Workshop and Summer School — August 2022
Simons Institute
Workshop Financial Assistance

The Mathematics of Machine Learning — May 2022
Women and Math Program
Institute for Advanced Study
Workshop Financial Assistance

Recent Developments on Mathematical/Statistical approaches in Data Science — June 2019
University of Texas, Dallas
Workshop Financial Assistance

Hackathon	Winner, Technical Track, Hacklytics Data Science at Georgia Tech	February 2019
Academic Awards and Scholarships	NITK Institute Gold Medal	2017
	1986 Batch Gold Medal	2017
	Prof. M. R. Shenoy Memorial Prize	2017
	Prof. K. M. Hebbar Gold Medal	2017
	NITK Surathkal Merit Scholarship	2013 - 2017
SERVICE	**Reviewer**, ICLR	2027
	Reviewer, NeurIPS	2026
	Reviewer, WiML Workshop at NeurIPS	2025
	Judge, AWM Poster Session at SIAM Annual Meeting	2025
	Mentor, AWM Workshop at SIAM Annual Meeting	2025
	Reviewer, Workshop Proposals for NeurIPS	2025
	Reviewer, WiML Workshop at NeurIPS	2024
	Reviewer/Volunteer, WiML workshop at NeurIPS	2023
	Volunteer, NeurIPS Conference	2023
	Volunteer, UAI Conference	2023
	Volunteer, WiML Workshop at ICML	2023
	Member, GT Mural Team	2022
	Reviewer, AIStats Conference	2021
	Reviewer, GT President's Undergraduate Research Award	2021 - 2022
	Panelist, GT ECE ORS Graduate Panel	2020
	Volunteer, GT ECE Prospective PhD student visit	2019 - 2022
	Teaching Volunteer, Shiksha, ACM NITK Student Chapter	2016
	Student Representative, NITK Student Council	2013 - 2014
TEACHING EXPERIENCE	**Mentor for undergraduate students**	
	Yue Teng	2020 - 2022
	Amran Mamuye, Eunsan Mo, Kerui Zhu, Robert Walker	2020 - 2021
	Guided the above students to work on a research project focused on simultaneous localization and mapping using paired comparisons of distances.	
	Graduate Teaching Assistant Georgia Tech ECE 6270 - Convex Optimization	Spring 2021
	Teaching Assistant Hands-on-Tech Georgia Tech Day Camp - Machine Learning	June 2019
	Graduate Teaching Assistant Georgia Tech CS 4641 - Machine Learning	Spring 2019
	Graduate Teaching Assistant Georgia Tech ECE 8843/ISYE 8843/CS 8803/BMED 8813 - Mathematical Foundations of Machine Learning	Fall 2018
RESEARCH EXPERIENCE	**Human Color Estimation Study** A user study was conducted where each participant was shown a target color and answered 20 sequential queries of the form 'Which among these two colors is closer to the target?'. We compare the performance of *confidence aware* response model against other response models. The estimation error, cross-entropy and accuracy of each model's predictions on the participant's choices, and mean self-reported confidence, averaged over all the participants, were compared. Our *confidence aware* model is	Fall 2026

the best on all four, and its queries receive the highest confidence, thereby proving it selects queries that balance both informativeness and response confidence.

Control parameter tuning using active synthesis Fall 2025 - present
Active preference learning was applied to optimize subjective controller gains for mobile robots. The framework dynamically synthesizes continuous-space queries to compare different trajectory parameters, asking users which control behavior they prefer. By translating this qualitative human feedback into a structured search, the method iteratively identifies optimal control parameters with minimal queries.

Active query synthesis for preference learning Fall 2022 - present
Proposed a *confidence aware* response model for pairwise comparisons, in which response reliability depends on the similarity of the compared options, to better capture human choices and improve preference learning. To avoid the computational cost of pool-based active learning, the *Info-Synth* query synthesis framework directly generates informative queries by maximizing mutual information in closed form, up to a one-dimensional search. For fixed datasets, the method employs tailored approximation strategies to select the closest available pairs to these synthesized queries, matching the sample efficiency of pool-based methods at lower computational cost.

Active learning for time series data Summer 2023
Developed a method for classifying individual instants of time series data. The data consists of features which repeat at regular intervals and the existence of these correlations poses a unique challenge for active label selection.

Unified framework for active learning Fall 2019 - Spring 2023
Developed a unified query framework for active learning based on nearest neighbor queries. This method can be applied to any problem which involves learning a representation of the dataset that reflects the underlying similarity. Demonstrated the performance of the method for active metric learning and active image classification using deep neural networks.

Active similarity learning and manifold graphs Fall 2019 - Spring 2023
Implemented active image classification strategies for semi-supervised classification on CIFAR-100 and DomainNet datasets. This project was funded by the *DARPA LwLL – Learning with Less labels* program.

Preference learning Fall 2019 - Summer 2022
The problem consists of estimating a user's preferences over a set of items. We use the *ideal point* model to localize a user in an embedding of items. Worked on a variety of problems involving localizing new items and users.

Switched Hawkes Processes Fall 2018 - Summer 2019
Developed the Switched Hawkes Process which can be used to model systems in which the parameters of the process dynamically change depending on some (known) external state. We propose a simple maximum likelihood estimation approach and apply our model to a real-world traffic sensor dataset to study traffic patterns during different configurations of the traffic lights at an intersection.

WORKSHOPS ATTENDED

Women in Data Science and Mathematics August 2023
Institute for Pure and Applied Mathematics
University of California, Los Angeles

Deep Learning Theory Workshop and Summer School August 2022
Simons Institute, University of California, Berkeley

The Mathematics of Machine Learning — May 2022
Women and Math Program
Institute for Advanced Study, Princeton, NJ

Algorithmic Learning Theory — March 2022
Mentoring Workshop
Online

Foundation of Data Science Summer School — August 2019
Georgia Institute of Technology, Atlanta, GA

Recent Developments on Mathematical/Statistical approaches in Data Science — June 2019
University of Texas at Dallas, Richardson, TX